

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE)

December 2012

CE402 Advanced Operating System

Date: 07.12.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Briefly explain the differences between user mode and kernel mode operation. Why two modes of operations are required in UNIX? [04]
- (b) What resources are used when a thread created? How do they differ from those when a process is created? [03]
- Q - 2 (a) List out the basic goals for Distributed system? Explain any two them in detail. [06]
- (b) What are vector clocks? What are the advantages of vector clock over Lamport clock? [04]
- (c) "Some processor do not support interrupt priorities in hardware." What are the alternatives? Explain in detail. [04]

OR

- Q - 2 (a) Give the differences: [06]
- (1) Distributed operating system Vs. Network operating system
- (2) Loosely-coupled systems Vs. Tightly-coupled systems
- (b) Why is computer clock synchronization necessary? Describe the design requirement for a system to synchronize the clocks in a distributed system? [04]
- (c) On a system with n distinct interrupt priority level, what is the maximum numbers of interrupts that may be nested at a time? What repercussions can this have on the sizes of various stacks? [04]
- Q - 3 (a) Suppose that two processes detect the demise of the coordinator simultaneously and both decide to hold an election using the bully algorithm. What will happen? Give a proper example. [06]
- (b) Which are the two basic approaches for distributed mutual exclusion? Explain any one approach in detail. [05]
- (c) How can we initialize, view and remove semaphore sets? [03]

OR

- Q - 3 (a) Ricart and Agrawala's algorithm has the problem that if a process has crashed and does not reply to a request from another process to enter a critical region, the lack of response will be interpreted as denial of permission. How would you address this problem? What mechanism would you suggest to differentiate between a denial of permission and a crashed process? [06]

- (b) What are the basic challenges for security in Distributed system? Explain any one of them in detail. [05]
- (c) "Kernel can directly access fields of the U area of the executing process but not of the U are of other process", Yes/No? Justify your answer. [03]

SECTION – II

- Q - 4 (a) **Attempt the following:** [04]
- (1) How does Asynchronous Transfer Mode (ATM) work?
 - (2) What is an idle workstation? How is an idle workstation found?
- (b) Differentiate sender-initiated and receiver-initiated Distributed heuristic algorithm. [03]
- Q - 5 (a) Briefly explain about RPC and describe design issues related to RPC in detail. [06]
- (b) What are the key features of distributed file systems? Explain any one of them in detail. [04]
- (c) Some operating systems provide support for concurrent processes, which may contain more than one thread. Such an operating system must provide a way for new processes to be created, and a way for new threads to be created within a process. Sketch, in words, what the OS must do to implement each of these operations. [04]

OR

- Q - 5 (a) How multiple memory modules and a scalable network allows the machine to scale to large number of processor? Explain with an example. [05]
- (b) What do you mean by Marshalling and Unmarshalling the data. Illustrate with an example. [05]
- (c) Differentiate Distributed memory systems and Shared memory system. [04]
- Q - 6 (a) **Answer the following (Any Three)** [12]
- (1) Explain Client-Server binding in RPC.
 - (2) Describe in detail atomicity and message ordering in group communication.
 - (3) In what scenario is stateful sever better than stateless server? Justify with proper example.
 - (4) What are basic requirements to the name structure and directory service?
- (b) What are the IPC functionality provided by pipes and message queues? [02]
